## Contact

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## Top Skills

Artificial Intelligence (AI)
Networking
Large Language Models (LLM)

## Languages

French (Limited Working)
Hindi (Native or Bilingual)
Marathi (Native or Bilingual)
English (Full Professional)
Kannada (Native or Bilingual)

## Certifications

Privacy Impact Assessment (PIA)
Fundatmentals of Generative AI
OAIC & Generative AI
Advanced C Programming: Pointers Master pointers, addresses and memory allocation in C
Australian Privacy Principles

## Honors-Awards

MRD Scholarship certificate
DAC
EDE Department Award
EDE2 Award
Sydney International Student Award

## Publications

Automated Script Evaluation using Machine Learning and Natural Language Processing
Eval - Automatic Evaluation of Answer Scripts using Deep Learning and Natural Language Processing

# Sagarika Manjunath Chavan

Masters of Computer Science @ The University of Sydney | Ex-Senior Engineer @ Bosch
Sydney, New South Wales, Australia

## Summary


I am currently pursuing a Master of Computer Science (Advanced Entry) at the University of Sydney, specializing in Artificial Intelligence, Data Science, and Cybersecurity. My research focuses on enhancing Large Language Model (LLM) agents for automated penetration testing using Reinforcement Learning, with the goal of improving the effectiveness and autonomy of cybersecurity assessments.Previously, I worked as a Senior Software Engineer at Bosch Global Software Technologies, contributing to Advanced Driver-Assistance Systems (ADAS) and data-driven solutions. More recently, I have worked as a Data Analyst Intern at Canopus Networks, where I was working on classifying network flows as research traffic. I have also worked as a Data Analytics and Machine Learning Intern at Cubic Innov8, developing AI-driven cybersecurity solutions for Security Operations Centers (SOCs) using Machine Learning and NLP techniques.I hold a Bachelor of Technology in Computer Science and Engineering with a specialization in Machine Intelligence and Data Science from PES University. My technical expertise spans AI, Machine Learning, Data Science, Cybersecurity, Data Engineering, Full-Stack Development, Cloud Computing, and Big Data technologies. Proficient in Python, Java, TypeScript, JavaScript, and modern AI/ML frameworks, I am passionate about applying technology to solve real-world challenges and continuously expanding my knowledge through research, innovation, and hands-on industry experience.


## Experience

Fiorin
Data Engineer
July 2026 - Present (4 months)
Sydney, New South Wales, Australia

## Canopus Networks

### Data Analyst

January 2026 - April 2026 (4 months)

Sydney, New South Wales, Australia

1. Developed an LLM-driven classification pipeline to identify research-related traffic from 3M+ encrypted network flow records using SNI/domain metadata.
2. Designed structured prompts and JSON-based outputs to label domains with research intent, confidence scores, and justification.
3. Implemented heuristic pre-filtering and post-processing validation to improve classification reliability at scale.
4. Analyzed behavioral flow metrics (uplink/downlink ratio, bytes per flow, duration) to characterize high-volume research data transfers.
5. Led development of the Supernetflow product website, articulating how Supernetflow provides deep, context rich network telemetry beyond traditional flow data to improve observability, traffic intelligence, and anomaly detection for enterprise networks.

## Cubic Innov8

### ML Intern

March 2025 - July 2025 (5 months)

Sydney, New South Wales, Australia

1. Spearheaded the development of a machine learning-based solution to automate remediation actions in Security Operations Centers (SOCs), using Microsoft’s GUIDE Security Incident Dataset with over 9.5 million records.
2. Built and evaluated three predictive models — Random Forest, XGBoost, and CatBoost — to classify incident severity (IncidentGrade) based on both structured features and NLP-processed alert text.
3. Applied advanced NLP techniques (TF-IDF, lemmatization, stopword removal) on unstructured alert titles to extract meaningful features, significantly improving model performance for minority classes.
4. Developed a real-time incident grade prediction pipeline to simulate SOC environments, enabling analysts to input raw alert titles and receive AI-driven triage suggestions instantly.
5. Tackled class imbalance using class weighting and stratified sampling, ensuring balanced learning across BenignPositive, TruePositive, and FalsePositive classes.
6. Delivered deep exploratory data analysis (EDA) using seaborn and matplotlib, uncovering key patterns in incident types, MITRE ATT&CK techniques, and grade distributions to guide modeling decisions.

7. Documented the full data science workflow, from data preprocessing and modeling to evaluation.

Bosch
Senior Engineer
August 2023 - February 2025 (1 year 7 months)
Bengaluru, Karnataka, India

1. Contributed to key areas of the ADAS (Advanced driver-assistance systems) PACE projects, advancing development and research initiatives to meet project goals.
2. Recognized for outstanding contributions and awarded a Certificate of Appreciation by the Department Head for my impactful performance in the EDE department.
3. I gained early career exposure through an onsite opportunity in Germany, where I collaborated with domain leads and cross-functional teams to align on project goals. Participated in a hackathon, contributing to smart suggestions and migrating key features from the backend to the frontend.
4. Contributed to three high-impact R&D tasks:
i. Search Bar Integration: Overcame technical challenges related to micro-frontend architecture by integrating a React Search Bar into the Angular application, also resolving shared state management issues.
ii. Video File Streaming: Developed an optimized solution to stream large video files by chunking data using a logarithmic algorithm, improving performance and reducing network requests.
iii. Marker Synchronization with Streaming Video: Developed mathematical logic to synchronize map markers with video timestamps, ensuring seamless integration between map data and streaming video.
5. Recognized for contributions to video file streaming and awarded at the EDE2 meeting for the project's timely completion.
6. Participated in the EDE Hackathon with the Royal Coders Bosch team, applying machine learning to minimize vector distances between models and ensuring backward compatibility. Leveraged Azure ML (AML) workspace for model training, inference, and management.
7. Actively organized team engagement activities during EDE events, fostering collaboration and morale. Assisted new joiners and interns with knowledge transfer sessions and created a comprehensive handbook to help them integrate into the team quickly.

Bosch
Project Trainee

January 2023 - May 2023 (5 months)
Bengaluru, Karnataka, India

Interned in the sector of autonomous driving.

Worked on multiple solutions as a Full Stack Developer, utilizing expertise in the frontend, backend, Docker, and cloud technologies. Contributed to the development of a tool that automates MLOps workflows and facilitates the anonymization of traffic-related images.

### PES University
Undergraduate Teaching Assistant
July 2022 - December 2022 (6 months)
Bengaluru, Karnataka, India

Undergraduate Teaching Assistant (TA) for the course of Big Data.
1) Assisted students in understanding and applying concepts related to Big Data.
2) Designed practical exercises for Apache Hive and conducted project viva sessions.
3) Delivered lectures on Hadoop MapReduce.

### Bosch Global Software Technologies
Project Trainee
June 2022 - July 2022 (2 months)
Bengaluru, Karnataka, India

Interned in the sector of autonomous driving.

Worked on visualizing data on map to understand data better and to measure its impact on business as raw data is converted to visual object.The main objective of the project was to Aggregate data using S2 cell concept and superclusters which provides robust constructive operations, representations of latitude - longitude points, geometric shapes over the unit sphere, robust algorithms etc.

### Cisco ThingQbator MNNIT
App Developer
July 2021 - December 2021 (6 months)
Bengaluru, Karnataka, India

Interned as a Flutter App Developer for Team Tregrantha. The team made it top 15 in the competition by Cisco and nasscom foundation.

### ISFCR and IoT Research Centre, PESU

IoT Intern

July 2021 - December 2021 (6 months)

Bangalore

ISFCR and IoT Research Centre, PESU

Developed a smartwatch project enabling wireless money transfer via Bluetooth with a unique proximity-based feature. Integrated a dedicated mobile app for wallet management, providing users a convenient solution for handling finances on the go.

## Education

### University of Sydney

Master's degree, Computer Science (Advanced Entry, Research Pathway) · (February 2025 - December 2026)

### PES University

Bachelor of Technology - BTech, Computer Science Engineering · (2019 - 2023)

### Deeksha, Main Campus, Bengaluru

12th, PUC · (2017 - 2019)

### Baldwin Girls' High School

10th, ICSE · (2017)